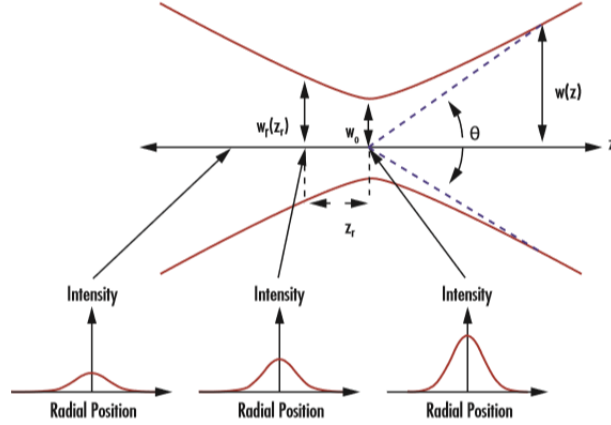


FSOC Data Rate calculations.
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Laser beam propagation¹



In the plot, $W_r(z_r)$ belongs to beam width at Rayleigh distance z_r , W_0 to the beam waist and $W(z)$ to the beam width at any distance z , and θ the beam divergence.

Divergence

$$\theta = \frac{\lambda}{\pi W_0}$$

Spot size

With $W_0 = w_0$, the beam spot size is given by:

$$w(z) \approx w_0 + z\theta$$

with differs from beam diameter, which is $2w(z)$ using this convention referred to the graphic.

Field of View (FOV)

$$\text{FOV} = 2\alpha = 2 \tan^{-1} \left(\frac{d}{2f} \right) \approx \frac{d}{f}$$

Gaussian beam

$$U_0(r, 0) = \exp \left(-\frac{r^2}{W_0^2} - \frac{ikr^2}{2F_0} \right)$$

$$U_0(r, L) = \frac{1}{p(L)} \exp(ikL) \exp \left(-\frac{r^2}{W_0^2} - \frac{ikr^2}{2F_0} \right)$$

$$p(L) = 1 - \frac{L}{F_0} + i \frac{2L}{kW_0^2}$$

¹most equations according to Andrew-Phillips book.

Gaussian beam parameters

$$\Theta_0 = 1 - \frac{L}{F_0}$$

$$\Lambda_0 = \frac{2L}{kW_0^2}$$

Gaussian Beam width

$$W = W_0 \sqrt{\Theta_0^2 + \Lambda_0^2}$$

Beam width at FWHM and $1/e^2$

$$W_{1/e^2} = \frac{W_{\text{FWHM}}}{\sqrt{2 \ln 2}}$$

Divergence at FWHM and $1/e^2$

$$\theta_{1/e^2} = \frac{2\lambda}{\pi W_0(0)}$$

$$\theta_{\text{FWHM}} = \sqrt{\frac{\ln 2}{2}} \cdot \theta_{1/e^2}$$

Irradiance on-axis

$$I(0, L) = I_0 \cdot \frac{W_0^2}{W^2} \left[\frac{\text{W}}{\text{m}^2} \right]$$

Power at the receiver

$$P_{R_x} = P_{T_x} \cdot \left(1 - e^{\frac{-2r^2}{w^2}} \right) [\text{W}]$$

Irradiance at the receiver

$$I_{R_x} = 4 \frac{P_{R_x}}{\pi D_{R_x}^2} \left[\frac{\text{W}}{\text{m}^2} \right]$$

Atmospheric channel effects

Kolmogorov outer scale and inner scale

Inertial range:

$$D_{RR}(R) = \langle (V_1 - V_2)^2 \rangle = C_V^2 R^{2/3}$$

$$D_{RR}(R) = \begin{cases} C_v^2 \ell_0^{-4/3} R^2 & , 0 < R \ll \ell_0 \\ C_v^2 R^{2/3} & , \ell_0 \ll R \ll L_0 \end{cases}$$

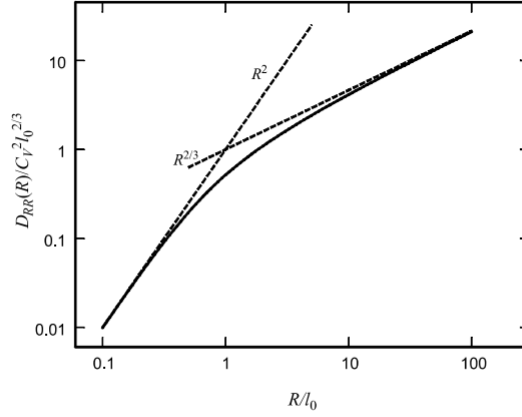


Figure 3.2 The scaled longitudinal velocity structure function showing both R^2 and $R^{2/3}$ asymptotic domains (dashed lines).

Beam spreading (Spot size)

Long-term spot size W_{LT} :

$$W_{LT}^2 = W^2 + W^2 T_{SS} + \langle r_c^2 \rangle = W_{ST}^2 + \langle r_c^2 \rangle$$

with $\langle r_c^2 \rangle$ the beam wander displacement variance.

Loss (Uplink)jki09:

$$L_{SR} = \left[1 + \left(\frac{D_T}{r_0} \right)^{\frac{5}{3}} \right]^{-\frac{6}{5}}$$

Beam wander

Beam wander variance:

$$\langle r_c^2 \rangle = 0.54 (H - h_{OGS})^2 \sec^2(\zeta) \left(\frac{\lambda}{2W_0} \right)^2 \left(\frac{2W_0}{r_0} \right)^{5/3}$$

RMS beam wander displacement:

$$\sqrt{\langle r_c^2 \rangle} = 0.69 \left(\frac{\lambda}{2W_0} \right) \left(\frac{2W_0}{r_0} \right)^{5/6}$$

with r_0 the Fried's parameter.

Refractive index structure parameter

HV: Hufnagel-Valley

$$C_n^2(h) = 0.00594 \left(\frac{w}{27} \right)^2 (10^{-5}h)^{10} \exp \left[-\frac{h}{1000} \right] + 2.7 \times 10^{-16} \exp \left[-\frac{h}{1500} \right] + A \exp \left[-\frac{h}{100} \right]$$

with $A = 1.17 \times 10^{-14} \text{m}^{-2/3}$ and $w = 21 \text{ m/s}$ recommended by ITU for night observations.

Rytov variance

$$\sigma_R^2 = 1.23 C_n^2 k^{7/6} L^{11/6}$$

Fried parameter (atmospheric coherence width)

For C_n^2 constant, horizontal propagation with uniform turbulence:

$$r_0 = (0.16 \cdot C_n^2 k^2 L)^{-3/5}$$

For Horizontal link (plane wave):

$$r_0 = (1.46 C_n^2 k^2 L)^{-3/5}$$

For Vertical link or slant paths:

$$r_0 = \left[0.42 \sec(\zeta) k^2 \int_{h_0}^H C_n^2(h) dh \right]^{-3/5}$$

with ζ the zenith angle ($\sec(0) = 1$), h_0 the height above ground level of the uplink transitter and/or downlink receiver, $H = h_0 + L \cos(\zeta)$ is the satellite altitude.

Downlink can be approximated by a plane wave. Use vertical link formula.

Uplink can be approximated by a spherical wave. Use next formula:

$$r_0 = \left[0.42 \sec(\zeta) k^2 \int_{h_0}^H C_n^2(h) \left(1 - \frac{h - h_{OGS}}{H - h_{OGS}} \right) dh \right]^{-3/5}$$

with H the satellite altitude, h_{OGS} the altitude of the OGS receiver, and h the integrand.

Greenwood frequency

$$f_g = 2.31 \lambda^{-6/5} \left[\sec(\theta) \int_{h_0}^H C_n^2(h) V_{\text{wind}}^{5/3}(h) dh \right]^{3/5}$$

Scintillation

Scintillation index:

$$\sigma_I^2 =$$

Loss:

$$L_{SI} = 4.343 \{ \operatorname{erf}^{-1}(2\rho_{th} - 1) [2 \ln(\sigma_I^2 + 1)]^{\frac{1}{2}} - \frac{1}{2} \ln(\sigma_I^2 + 1) \}$$

Data Communication

Baud Rate

Bandwidth

Modulation

Spectral Efficiency

Capacity

$$C = B \cdot \log_2 \left(1 + \frac{S}{N} \right)$$

1 Optical Communication

Signal-to-Noise Ratio

$$\text{SNR}_0 = \frac{i_s}{\sigma_N} = \sqrt{\frac{\eta P_s}{2h\nu B}} \rightarrow 10 \cdot \log(\text{SNR}_0) = 74.9 \text{ dB}$$

$$i_s = \frac{\eta e P_s}{h\nu} = \frac{0.85 \cdot e \cdot 10 \text{ mW}}{h \cdot 193.4 \text{ THz}} = 0.01063$$

$$\langle i_N^2 \rangle = 2eBi_s = 2 \cdot e \cdot 100 \cdot 10^9 \cdot 0.01063 = 3.406 \cdot 10^{-10}$$

Bit Error Rate

$$4\text{-QAM: } BER \approx \cdot Q \left(\sqrt{\frac{2E_b}{N_0}} \right)$$

$$16\text{-QAM: } BER \approx 3 \cdot Q \left(\sqrt{\frac{4E_b}{5N_0}} \right)$$

Mean SNR

$$\langle \text{SNR} \rangle = \frac{\text{SNR}_0}{\sqrt{\left(\frac{P_{s0}}{\langle P_s \rangle}\right) + \sigma_I^2(D_G)\text{SNR}_0^2}}$$

$$\langle \text{SNR} \rangle \cong \frac{1}{\sigma_1(D_G)}, \quad \text{SNR}_0 \rightarrow \infty$$

$$\langle \text{SNR} \rangle = \frac{\langle i_S \rangle}{\sigma_N}$$

Fading statistics

$$Pr_{\text{fade}} = 1 - \frac{1}{2} \int_0^\infty (u) \text{erfc} \left(\frac{\text{TNR} - \langle \text{SNR} \rangle u}{\sqrt{2}} \right) du$$

$$Pr(E) = \langle \text{BER} \rangle = \frac{1}{2} \int_0^\infty p_f(u) \text{erfc} \left(\frac{\langle \text{SNR} \rangle u}{2\sqrt{2}} \right) du$$

2 Modulation

OOK: On-Off Keying

QPSK: Quadrature Phase Keying

QAM: Quadrature Amplitude Modulation

Data Rates

Photodetectors

Noise

APD

PIN

InGaAS

Example: InGaAs PIN photodiode G6854-01

- Sensitivity $S = 0.95 \text{ A/W}$
- Noise Equivalent Power: $\text{NEP} = 2 \times 10^{-15} \text{ W}/\sqrt{\text{Hz}}$

QPD

Laser transmitter

Amplitude modulation

Phase modulation

Polarization modulation

3 References

?? <https://www.nsnam.org/reviews/2016/socis-midterm/fso.html>